观测方程: $P_R^s=\sqrt{(X_i^s-X_0)^2+(Y_i^s-Y_0)^2+(Z_i^s-Z_0)^2}+\delta t_{R_i}+\Delta_i$

线性化: $P_R^s= D_i-\frac{X_i-X_0}{D_i}dx_0-\frac{y_i-y_0}{D_i}dy_0-\frac{z_i-z_0}{D_i}dz_0+\delta t_{R0}+\Delta_i$ — $\Delta P-P_0 = dx\ \ dy\ \ dz+dtr+d\varepsilon$

$$=\underset{n*1}{P}=\underset{n*(n+4)}{\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & & & & \\ -\frac{x_1-x_0}{D_1} & -\frac{y_1-y_0}{D_1} & -\frac{z_1-z_0}{D_1} & 1 & 1 & 0 & \cdots & 0 & 0 \\ & & & & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & & \ddots & & \vdots \\ & & & & 0 & 0 & \cdots & 1 & 0 \\ -\frac{x_n-x_0}{D_n} & -\frac{y_n-y_0}{D_n} & -\frac{z_n-z_0}{D_n} & 1 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}}\underset{n+4*1}{\begin{bmatrix} dx_0\\ dy_0\\ dz_0\\ \delta t_{R0}\\ \Delta_1\\ \vdots\\ \Delta_n\end{bmatrix}}+\underset{n*1}{\begin{bmatrix} D_1\\ D_2\\ \vdots\\ D_n\end{bmatrix}}=\underbrace{\begin{bmatrix} -\frac{x_1-x_0}{D_1} & -\frac{y_1-y_0}{D_1} & -\frac{z_1-z_0}{D_1}\\ \vdots & \vdots & \vdots\\ -\frac{x_n-x_0}{D_n} & -\frac{y_n-y_0}{D_n} & -\frac{z_n-z_0}{D_n}\end{bmatrix}\begin{bmatrix} dx\\ dy\\ dz\end{bmatrix}+\begin{bmatrix}\Delta_1\\ \vdots\\ \Delta_i\end{bmatrix}}_{E=0}+\begin{bmatrix}\delta t_{r0}\\ \vdots\\ \delta t_{r0}\end{bmatrix}+\begin{bmatrix} D_1\\ \vdots\\ D_n\end{bmatrix}$$

$$\Downarrow$$

$$X=\underset{4*(n+4)}{\begin{bmatrix} 1 & & & & 0 & \cdots & 0\\ & 1 & & & \vdots & \ddots & \vdots\\ & & 1 & & & & \\ & & & 1 & 0 & \cdots & 0\\ & \underset{4\times4}{} & & & & \underset{4\times n}{} & \end{bmatrix}}\begin{bmatrix} dx\\ dy\\ dz\\ d\delta t\\ \Delta_1\\ \vdots\\ \Delta_n\end{bmatrix}=B_2M \qquad \underset{n*1}{\mu_p}=\begin{bmatrix}\delta t_{r0}\\ \vdots\\ \delta t_{r0}\end{bmatrix}+\begin{bmatrix} D_1\\ \vdots\\ D_n\end{bmatrix}$$

$$Q=\underset{(n+4)*(n+4)}{\begin{bmatrix} 10 & & & & & & \\ & 10 & & & & & \\ & & 10 & & & & \\ & & & 10 & & & \\ & & & & obs_1[3] & & \\ & & & & & obs_2[3] & \\ & & & & & & \ddots\ obs_n[3]\end{bmatrix}}$$

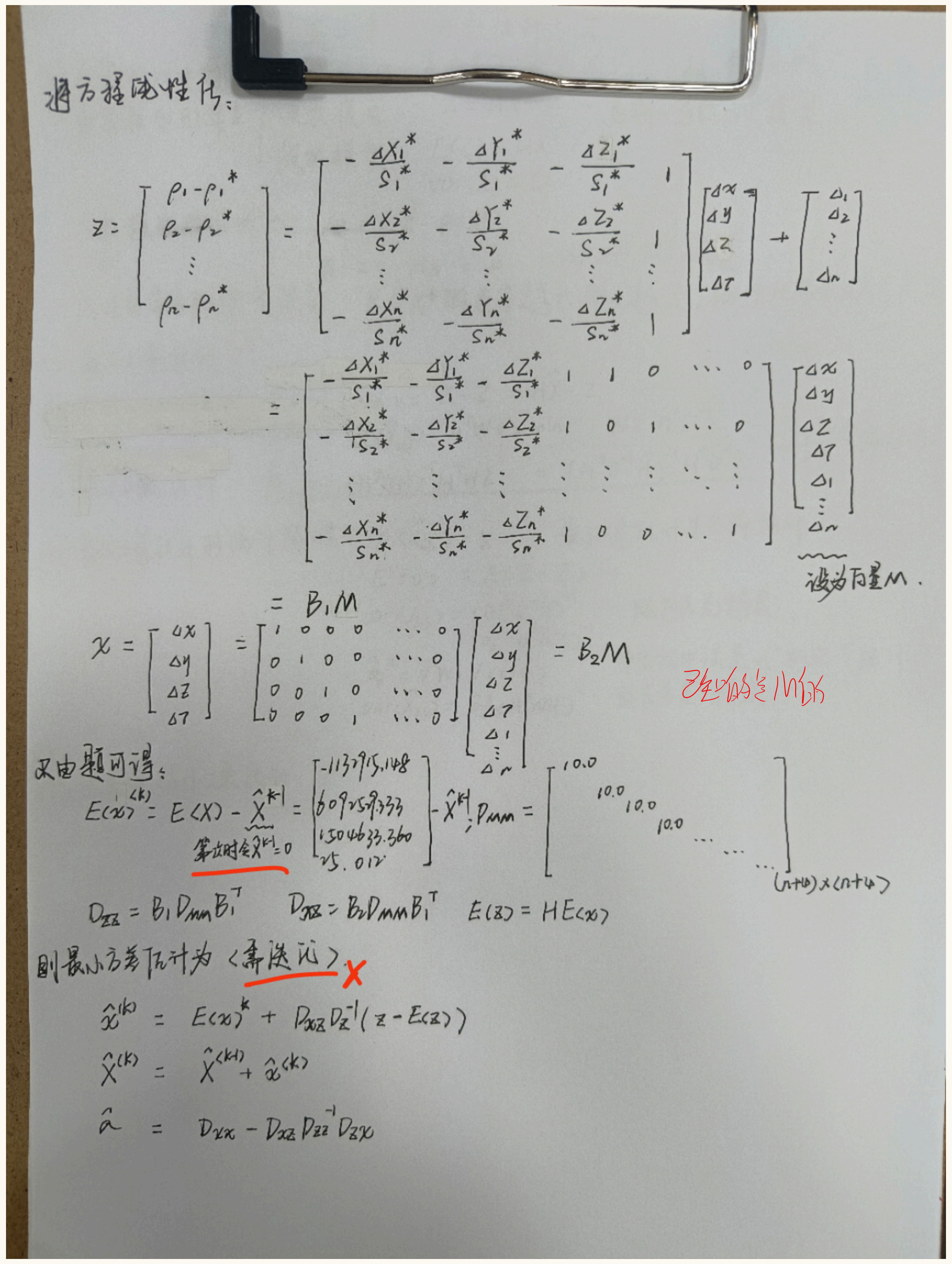
$$\underset{n*n}{D_p}=BQB^T$$

$$=\underset{n*(n+4)}{\begin{bmatrix} -\frac{x_1-x_0}{D_1} & -\frac{y_1-y_0}{D_1} & -\frac{z_1-z_0}{D_1} & 1 & 1 & 0 & \cdots & 0 & 0\\ & & & & 0 & 1 & \cdots & 0 & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & & \ddots & & \vdots\\ & & & & 0 & 0 & \cdots & 1 & 0\\ -\frac{x_n-x_0}{D_n} & -\frac{y_n-y_0}{D_n} & -\frac{z_n-z_0}{D_n} & 1 & 0 & 0 & \cdots & 0 & 1\end{bmatrix}}\underset{(n+4)*(n+4)}{\begin{bmatrix} 10 & & & & & & \\ & 10 & & & & & \\ & & 10 & & & & \\ & & & 10 & & & \\ & & & & obs_1[3] & & \\ & & & & & obs_2[3] & \\ & & & & & & \ddots\ obs_n[3]\end{bmatrix}}\underset{(n+4)*n}{\begin{bmatrix} -\frac{x_1-x_0}{D_1} & -\frac{y_1-y_0}{D_1} & -\frac{z_1-z_0}{D_1} & 1 & 1 & 0 & \cdots & 0 & 0\\ & & & & 0 & 1 & \cdots & 0 & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & & \ddots & & \vdots\\ & & & & 0 & 0 & \cdots & 1 & 0\\ -\frac{x_n-x_0}{D_n} & -\frac{y_n-y_0}{D_n} & -\frac{z_n-z_0}{D_n} & 1 & 0 & 0 & \cdots & 0 & 1\end{bmatrix}^T}$$

$$\underset{4*n}{D_{xp}}=\underset{4*(n+4)}{\begin{bmatrix} 1 & & & & 0 & \cdots & 0\\ & 1 & & & \vdots & \ddots & \vdots\\ & & 1 & & & & \\ & & & 1 & 0 & \cdots & 0\end{bmatrix}}\underset{(n+4)*(n+4)}{\begin{bmatrix} 10 & & & & & & \\ & 10 & & & & & \\ & & 10 & & & & \\ & & & 10 & & & \\ & & & & obs_1[3] & & \\ & & & & & obs_2[3] & \\ & & & & & & \ddots\ obs_n[3]\end{bmatrix}}\underset{(n+4)*n}{\begin{bmatrix} -\frac{x_1-x_0}{D_1} & -\frac{y_1-y_0}{D_1} & -\frac{z_1-z_0}{D_1} & 1 & 1 & 0 & \cdots & 0 & 0\\ & & & & 0 & 1 & \cdots & 0 & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & & \ddots & & \vdots\\ & & & & 0 & 0 & \cdots & 1 & 0\\ -\frac{x_n-x_0}{D_n} & -\frac{y_n-y_0}{D_n} & -\frac{z_n-z_0}{D_n} & 1 & 0 & 0 & \cdots & 0 & 1\end{bmatrix}^T}=B_2QB_1$$

$$\underset{4*1}{\hat{X}_{mv}}=\underset{4*1}{\mu_x}+\underset{4*n}{D_{xp}}\underset{n*n}{D_p^{-1}}(\underset{n*1}{p-\mu_p})$$

$$Var(\hat{X}_L)=\underset{4\times4}{D_x}-\underset{4n}{D_{xp}}\underset{n\times n}{D_p^{-1}}\underset{n\times4}{D_{px}}$$



误方程线性化:

$$z=\begin{bmatrix}\rho_1-\rho_1^*\\ \rho_2-\rho_2^*\\ \vdots\\ \rho_n-\rho_n^*\end{bmatrix}=\begin{bmatrix}-\frac{\Delta X_1^*}{S_1^*} & -\frac{\Delta Y_1^*}{S_1^*} & -\frac{\Delta Z_1^*}{S_1^*} & 1\\ -\frac{\Delta X_2^*}{S_2^*} & -\frac{\Delta Y_2^*}{S_2^*} & -\frac{\Delta Z_2^*}{S_2^*} & 1\\ \vdots & \vdots & \vdots & \vdots\\ -\frac{\Delta X_n^*}{S_n^*} & -\frac{\Delta Y_n^*}{S_n^*} & -\frac{\Delta Z_n^*}{S_n^*} & 1\end{bmatrix}\begin{bmatrix}\Delta x\\ \Delta y\\ \Delta z\\ \Delta\tau\end{bmatrix}+\begin{bmatrix}\Delta_1\\ \Delta_2\\ \vdots\\ \Delta_n\end{bmatrix}$$

$$=\begin{bmatrix}-\frac{\Delta X_1^*}{S_1^*} & -\frac{\Delta Y_1^*}{S_1^*} & -\frac{\Delta Z_1^*}{S_1^*} & 1 & 1 & 0 & \cdots & 0\\ -\frac{\Delta X_2^*}{S_2^*} & -\frac{\Delta Y_2^*}{S_2^*} & -\frac{\Delta Z_2^*}{S_2^*} & 1 & 0 & 1 & \cdots & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots\\ -\frac{\Delta X_n^*}{S_n^*} & -\frac{\Delta Y_n^*}{S_n^*} & -\frac{\Delta Z_n^*}{S_n^*} & 1 & 0 & 0 & \cdots & 1\end{bmatrix}\underbrace{\begin{bmatrix}\Delta x\\ \Delta y\\ \Delta z\\ \Delta\tau\\ \Delta_1\\ \vdots\\ \Delta_n\end{bmatrix}}_{\text{设为向量}M.}$$

$$=B_1M$$

$$x=\begin{bmatrix}\Delta x\\ \Delta y\\ \Delta z\\ \Delta\tau\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0 & \cdots & 0\\ 0 & 1 & 0 & 0 & \cdots & 0\\ 0 & 0 & 1 & 0 & \cdots & 0\\ 0 & 0 & 0 & 1 & \cdots & 0\end{bmatrix}\begin{bmatrix}\Delta x\\ \Delta y\\ \Delta z\\ \Delta\tau\\ \Delta_1\\ \vdots\\ \Delta_n\end{bmatrix}=B_2M$$

又由题可得:

$$E(x)^{(k)}=E(X)-\underset{\text{第一次时令}\hat{X}^{k-1}=0}{\hat{X}^{k-1}}=\begin{bmatrix}-1137915.148\\ 6092579.333\\ 1504633.360\\ 25.012\end{bmatrix}-\hat{X}^{k-1};\quad D_{MM}=\underset{(n+4)\times(n+4)}{\begin{bmatrix}10.0 & & & & & & \\ & 10.0 & & & & & \\ & & 10.0 & & & & \\ & & & 10.0 & & & \\ & & & & \cdots & & \\ & & & & & \cdots & \\ & & & & & & \cdots\end{bmatrix}}$$

$D_{zz}=B_1D_{MM}B_1^T \qquad D_{xz}=B_2D_{MM}B_1^T \qquad E(z)=HE(x)$

则最小方差估计为（非迭代）✗

$$\hat{x}^{(k)}=E(x)^k+D_{xz}D_z^{-1}(z-E(z))$$

$$\hat{X}^{(k)}=\hat{X}^{(k-1)}+\hat{x}^{(k)}$$

$$\hat{\alpha}=D_{xx}-D_{xz}D_{zz}^{-1}D_{zx}$$

迭代的是小误

一步预测

状态观测 $\underset{4\times1}{\hat{X}(k,k-1)}=\underset{4\times4}{\Phi_{k,k-1}}\underset{4\times1}{\hat{X}(k-1)} \qquad \Phi_{k,k-1}=I$

预测方差 $\underset{4\times4}{D_{\hat{X}}(k,k-1)}=\phi_{k,k-1}D_{\hat{X}}(k-1)\phi_{k,k-1}^T+D_w(k-1)$

$=\underset{4\times4}{D_{\hat{X}}(k-1)}+\underset{4\times4}{D_w(k-1)}$

$\underset{4\times5}{\hat{X}_0}=[x_0\ y_0\ z_0\ t_0]^T$（观测均值）

$$\underset{4\times4}{D_{X_0}}=\begin{bmatrix}100 & & & \\ & 100 & & \\ & & 100 & \\ & & & 100\end{bmatrix} \qquad D_w(k-1)=$$

测量更新 K

$$z=-\frac{x_i-x_0}{D_i}dx_0-\frac{y_i-y_0}{D_i}dy_0-\frac{z_i-z_0}{D_i}dz_0+\delta t_{R0}+\Delta_i$$

$$=HX+\Delta$$

$$H_k=\underset{n\times4}{\begin{bmatrix}-\frac{x_1-x_0}{D} & -\frac{y_1-y_0}{D} & -\frac{z_1-z_0}{D} & 1\\ \vdots & \vdots & \vdots & \vdots\\ -\frac{x_n-x_0}{D} & -\frac{y_n-y_0}{D} & -\frac{z_n-z_0}{D} & 1\end{bmatrix}} \qquad \Delta:$$

$$\underset{4\times n}{K_k}=\underset{4\times4}{D_{\hat{X}}(k,k-1)}\underset{4\times n}{H_k^T}(\underset{n\times4}{H_k}\underset{4\times4}{D_{\hat{X}}(k,k-1)}\underset{4\times n}{H_k^T}+\underset{n\times n}{D_\Delta(k)})^{-1}$$

$$\underset{n\times1}{V_{k,k-1}}=\underset{n\times1}{Z(k)}-\underset{n\times4}{H_k}\underset{4\times1}{\hat{X}(k,k-1)}$$

$$\underset{4\times1}{\hat{X}(k)}=\hat{X}(k,k-1)+\Delta X=\underset{4\times1}{\hat{X}(k,k-1)}+\underset{4\times n}{K_k}\underset{n\times1}{V(k,k-1)}$$

滤波方差:

$$\hat{X}(k,k-1)=\phi_{k,k-1}\hat{X}(k-1)$$

一步预测 抛层卡

$$\underset{4\times1}{X(t)}=\underset{4\times4}{A(t)}\underset{4\times1}{X(t)}+\underset{4\times1}{G(t)}+\underset{4\times2}{C(t)}\underset{2\times1}{e(t)}$$

$$\downarrow\qquad\downarrow\qquad\downarrow\qquad\downarrow$$

$$\begin{bmatrix}v_x\\ -k_xv_x^2+e_x\\ v_y\\ k_yv_y^2-g+e_y\end{bmatrix}\quad\begin{bmatrix}x\\ v_x\\ y\\ v_y\end{bmatrix}\quad\begin{bmatrix}0 & 0\\ 1 & 0\\ 0 & 0\\ 0 & 1\end{bmatrix}\quad\begin{bmatrix}e_x\\ e_y\end{bmatrix}$$

$$\underset{4\times4}{A(t)}=\begin{bmatrix}0 & 1 & & \\ & -2k_xv_x & 0 & \\ & & 0 & 1\\ & & & 2k_yv_y\end{bmatrix}\ 有 \qquad \underset{4\times1}{G(t)}=\begin{bmatrix}0\\ k_xv_x(t)\\ 0\\ -k_yv_y^2-g\end{bmatrix}$$

微分方程: $X(k,k-1)=\underset{4\times4}{\phi}X(k-1)+\underset{4\times1}{\Omega(k-1)}+W(k-1)$

ϕ → 离散 $I+A(t-\tau)$；Ω → $\int\phi G(t)d\tau$；W → $\int\phi C e(\tau)d\tau$

$=\int_{t_{k-1}}^{t_k}\phi(t_k,\tau)d\tau\cdot G \qquad =0$

$\phi=有$

$\Omega=有$

$$\underset{4\times4}{D_w(k-1)}=\int_{t_{k-1}}^{t_k}\phi(t_k,\tau)CD_e(\tau)C^T\phi(t_k,\tau)d\tau\ =有$$

$$\downarrow$$

$$\underset{4\times4}{D_{\hat{X}}(k,k-1)}=\underset{4\times4}{\phi}\underset{4\times4}{D_{\hat{X}}(k-1)}\underset{4\times4}{\phi}+\underset{4\times4}{D_w(k-1)}\qquad =$$

测量更新:

观测值逐次更新: 观测方程: $r(k)=$

2个题2个方向: $\alpha(k)=$

$$\underset{2\times2}{D_\Delta(k)}=\begin{bmatrix}10\,m^2 & \\ & 1\times10^{-5}\,rad^2\end{bmatrix}$$

r:

$V_r=r(k)-\sqrt{x^{0\,2}+y^{0\,2}}$

$\underset{1\times4}{h_r}=[\ \ 0\ \ \ 0]$ 线性化

$$\underset{4\times1}{K_{rk}}=\frac{\underset{4\times4}{D_{\hat{X}}(k,k-1)}\underset{4\times1}{h_r(k)^T}}{h_r(k)D_{\hat{X}}(k,k-1)h_r(k)^T+d_{\Delta r}(k))}$$

$$\underset{4\times1}{\hat{X}(k)}=\underset{4\times1}{\hat{X}(k,k-1)}+\underset{4\times1}{K_{rk}}\underset{1\times1}{V_r(k,k-1)}$$

$$\underset{4\times1}{D_{\hat{X}}(k)}=(\underset{4\times4}{I}-\underset{4\times1}{K_{rk}}\underset{1\times4}{h_r(k)})\underset{4\times4}{D_{\hat{X}}(k,k-1)}$$

α:

$V_\alpha=\alpha(k)-\sqrt{x^{0\,2}+y^{0\,2}}$

$h_\alpha=[\ \ 0\ \ \ 0]$ 线性化

$$K_{\alpha k}=\frac{D_{\hat{X}}(k,k-1)h_r(k)^T}{h_r(k)D_{\hat{X}}(k,k-1)h_r(k)^T+d_{\Delta r}(k))}$$

$$\hat{X}(k)=\hat{X}(k,k-1)+K_{rk}V_r(k,k-1)$$

$$D_{\hat{X}}(k)=(I-K_{rk}h_r(k))D_{\hat{X}}(k,k-1)$$